



Statistics of Financial Markets

Chapter 15: Review



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Outline

Course Summary

Key Concepts

Exam Preparation

References

Course Summary

- ▣ Overview of all topics covered in Chapters 1–14
- ▣ Efficient Market Hypothesis and random walk models
- ▣ Variance ratio tests and stylized facts of financial returns
- ▣ Volatility modeling: ARCH, GARCH, and asymmetric extensions
- ▣ Risk measures: VaR and Expected Shortfall
- ▣ Credit scoring and machine learning in finance
- ▣ Cryptocurrency markets and stablecoins
- ▣ Systemic risk measurement and network analysis
- ▣ Fractal Market Hypothesis: Hurst exponent and long memory

Key Concepts

- ▣ Market efficiency and its three forms
- ▣ Random walk types (RW1, RW2, RW3) and their tests
- ▣ Stylized facts: heavy tails, volatility clustering, leverage effect
- ▣ GARCH family: specification, estimation, model selection
- ▣ VaR estimation methods and backtesting
- ▣ Logistic regression and ROC/AUC for scoring models
- ▣ LASSO, Random Forest, and model validation
- ▣ CoVaR, FRM, and systemic risk networks
- ▣ Fractal Market Hypothesis, R/S analysis, and DFA

Exam Preparation

- Exam format and structure
- Key definitions and theorems to review
- Practice problems and worked examples
- Tips for computational questions (Python/R)
- Frequently asked questions from previous years

References

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). Quantitative Risk Management.
- ▣ Tsay, R.S. (2010). Analysis of Financial Time Series.